

Cooling cravers to the carefree: Behavioral archetypes of thermal preference from city-scale longitudinal smartwatch micro-surveys

Clayton Miller^{a,*}

^aSingapore Management University, *TODO(author): department or school, Singapore, TODO(author): postal code, Singapore*

Abstract

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1. Introduction

Characterization of thermal comfort in the built environment has long been dominated by the creation of thermal comfort models, which combine physiology and environmental science to describe when and how a person should feel thermally comfortable or uncomfortable [1]. Rooted in heat-balance theory, these models relate metabolic heat production, clothing insulation, air temperature, radiant conditions, humidity, and air movement to a predicted comfort state, drawing on how core and skin temperatures jointly drive thermal sensation [2], how circadian phase and the thermoneutral zone set the body's baseline [3, 4], and how radiant and solar exposure adds heat that simple air-temperature metrics miss [5, 6]. From these inputs, the models predict thermal sensation, satisfaction, and preference, and they remain the backbone of the comfort standards and mechanical systems designed around them [7]. Newer adaptive formulations extend this picture by recognizing that people stay comfortable across a wider range of conditions than heat-balance theory alone would predict, as behavioral adjustment, expectation, and prior thermal history reshape what feels acceptable [8, 9]. Even within the same space, individuals differ systematically in thermal sensitivity and response [10, 11], which is one reason identical ambient conditions can produce the heterogeneous preferences illustrated in Figure 1. To capture this variation directly, personal comfort models learn from an individual's own repeated feedback and physiological signals rather than assuming a population average [12]. Yet both approaches face practical limits. Generalized models predict individual thermal sensation correctly only about a third of the time [7], while personal models are difficult to scale because they depend on collecting sustained, labeled data from every occupant [12]. Efforts to connect these models to the broader context that surrounds them, including the psychological and non-physical influences that shape perception [13, 14], demographic and cultural differences in expectation and preference

[15, 16], and the social conventions and economics that govern who controls and pays for indoor climate [17, 18], remain nascent, but they may explain much of why the models still fall short.

1.1. Thermal behavior varies between people under similar conditions

Given these deficiencies, it is worth looking more closely at the wider set of factors that shape people's thermal experience in cities, which Figure 1 organizes into two broad families. The left side gathers the physical, environmental, and physiological factors that classical models were built to handle but that continue to surprise in their detail and interaction. At the environmental scale, real spaces are rarely uniform, and local asymmetries in radiant, surface, and air-movement conditions shape comfort in ways a single room temperature cannot represent [19], while personally controlled air movement can restore comfort and even cognitive performance at warmer set points [20, 21]. Closer to the body, near-body heat sources such as laptops [22], the insulation and cut of clothing [23], posture [24], and physical activity and metabolic rate [25] all shift the balance between heat production and loss. The physiological state of the occupant adds further variation through cutaneous blood flow [26], sweating and skin wettedness [27], hydration [28], the thermic effect of food and drink [29, 30], and age-related shifts in thermal thresholds [31]. Even the pleasantness of a given thermal stimulus depends on internal state, so the same temperature can feel welcome or aversive depending on whether it helps restore the body's balance [32]. These physical and physiological signals also rarely act alone, combining with the visual, acoustic, and air-quality exposures that any rigorous account of the indoor experience must weigh together [33].

The right side of Figure 1 gathers the psychological, cultural, economic, and historical or institutional forces that sit outside the heat balance yet strongly condition how people interpret and act on it. Psychologically, cooling choices are bound up with identity, values, and the situation at hand rather than temperature alone [34], and habitual reliance on air conditioning can harden into a practice that people find difficult to step outside of, even briefly [35]. Culturally, expectations of what

*Corresponding author
Email address: *TODO(author)*:
corresponding_email@example.edu (Clayton Miller)

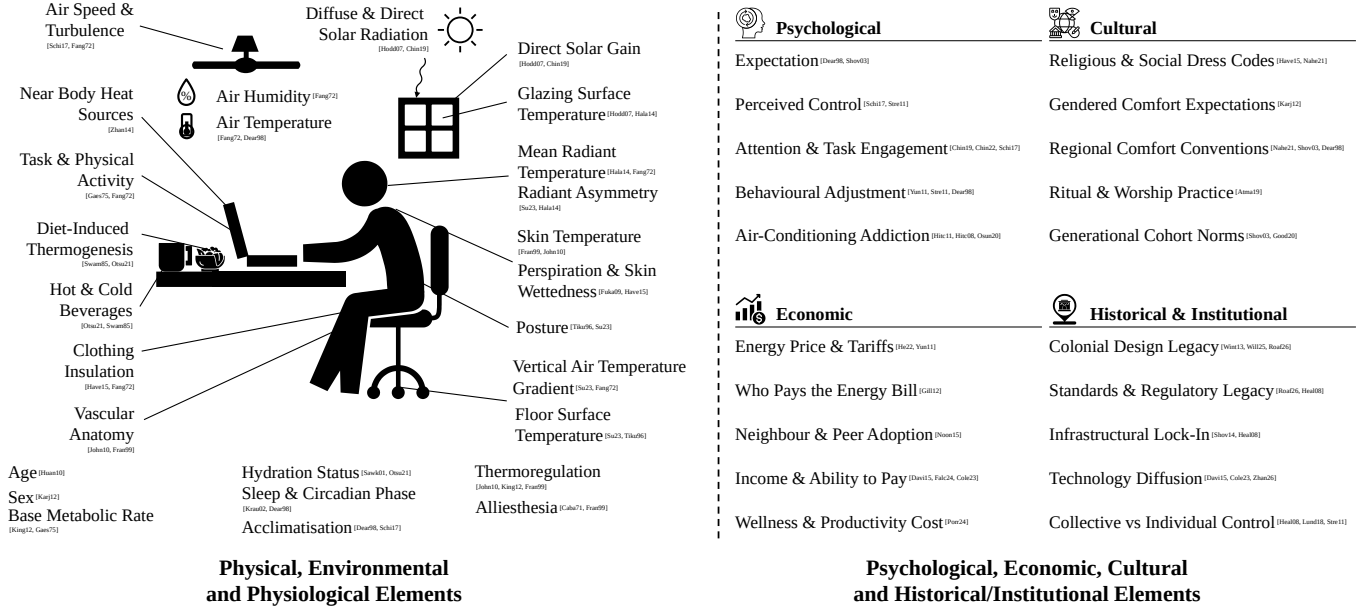


Figure 1: Elements shaping individual thermal preference. The left side groups physical, environmental, and physiological factors around a seated occupant: the surrounding radiant, air-movement, and surface conditions; near-body heat sources, clothing insulation, posture, and physical activity; and physiological state such as skin temperature, perspiration, metabolic rate, hydration, sleep and circadian phase, acclimatization, and thermoregulation. The right side lists psychological, cultural, economic, and historical or institutional factors, ranging from expectation, perceived control, and behavioral adjustment, to dress conventions and regional comfort norms, energy prices and who pays the bill, and colonial design, regulatory, and infrastructural legacies. Together these illustrate why identical ambient conditions can produce heterogeneous thermal-preference responses across individuals.

a comfortable indoor climate should feel like are learned and socially organized, shaped by evolving conventions, domestic technology, and the material culture of cooling [36, 37, 38], and they take on distinctive forms in specific settings, from the air-conditioned lives of young people in Singapore [39] to the use and energy demand of naturally ventilated places of worship [40]. Economically, the decision to buy and run cooling responds to weather, prices, incomes, and the efficiency of the available equipment [41, 42, 43], and as incomes rise and the climate warms, adoption is projected to grow steeply, with large and unequally distributed consequences for electricity demand and emissions [44, 45, 46, 47]. Underlying all of this is a historical and institutional legacy in which particular comfort norms, standards, and infrastructures were exported and standardized, homogenizing expectations across very different climates and cultures [48, 49, 50, 51], a trajectory whose sustainability in an increasingly hot world is now openly in question [52].

1.2. Repeated in-situ data can reveal behavioral response patterns

Capturing the convergence of these physical and non-physical influences is difficult in controlled or laboratory settings, where the very context that shapes perception is stripped away. Observing people in their natural environments over time is one of the few ways to see how thermal experience unfolds across the varied spaces, activities, and social situations of daily life. Repeated in-situ reporting through ecological momentary assessment provides a practical route to these observations, recording subjective states as they occur rather

than in recollection [53]. The data analyzed here come from exactly such a deployment, collected in Singapore with the open-source Cozie smartwatch platform, which delivers micro-surveys directly on an Apple Watch and pairs each response with the wearer’s onboard physiological signals and linked outdoor weather [54]. The deployment grew out of a broader program that used smartwatch-based just-in-time adaptive interventions to nudge heat and noise mitigation in the urban environment [55, 56], and the resulting dataset has since supported a public machine-learning competition on predicting heat and noise perception from digital-twin context [57, 58] as well as related work linking wearable reports to urban digital twins [59, 60]. Previous research characterizes this same deployment of 9,962 geolocated micro-survey responses from 106 participants, showing that everyday urban spaces separate most sharply along behavioral and acoustic lines and that thermal preference tracks the immediate controlled environment more than the outdoor weather [61]. That space-level characterization motivates the present study, which expands the behavioral dimension of the same dataset by shifting the unit of analysis from the space to the person. Rather than asking how perception differs between spaces, this study clusters participants by their repeated thermal-preference behavior to uncover person-level archetypes, and then examines whether participant meta-data offers clues to the non-physical and environmental elements of their thermal experience, with the analytical sample and eligibility screen detailed in Section 2.2.

2. Methods

This method builds directly on a smartwatch micro-survey deployment that was developed and reported across several prior studies, summarized here before detailing the archetype analysis that is new to this work. The data were collected in Singapore with the open-source Cozie platform, an Apple Watch application that administers short micro-surveys and links each response to the wearer’s onboard physiological signals and to matched outdoor weather [54]. The deployment itself grew out of a smartwatch-driven just-in-time adaptive intervention framework for building occupants [55], and its instrumentation, recruitment, and preliminary results have been described in the methods and competition papers that accompanied it [57, 58]. Participants were recruited in and around the National University of Singapore under approval from its Institutional Review Board, gave informed consent, and wore a paired Apple Watch and iPhone during working hours for approximately four weeks while completing repeated micro-surveys [61]. The resulting privacy-protected dataset of 9,962 geolocated responses from 106 participants was released publicly for a machine-learning competition [57, 58] and is characterized at the level of urban space in the companion paper [61]. Within this dataset, the 9,520 responses that carry both a thermal-preference answer and a matched outdoor heat index form the basis for the present analysis, with a median of 96 responses per participant across the cohort. Reported preferences skew toward no change, cooler preferences are far more common than warmer ones, most responses come from the home, and the matched outdoor heat index is high and narrow around a median near 34 °C, consistent with the warm tropical setting and characterized more fully in the companion paper [61]. Here the same underlying deployment is reused but analyzed at the level of the person, and the subsections below describe the survey measures and environmental-control context, the participant eligibility screen and person-level features, the archetype-discovery procedure, and the robustness checks used to scrutinize the resulting groupings.

2.1. Survey measures, outdoor heat, and environmental-control context

At each submission, the Cozie micro-survey presents a short battery of questions that records the participant’s thermal preference and the level of nearby noise, together with conditional follow-ups on the kind of noise and earphone use, the location or space type, whether the participant is *Alone* or in a *Group*, and the activity being performed [61]. Figure 2 shows the two survey questions that this analysis uses as its primary components, thermal preference and location. The remainder of the battery, along with the linked physiological, weather, and urban-context data, is treated as outside the scope of this study. The thermal-preference question asks whether a participant would prefer to be *Cooler*, feel *No change*, or prefer to be *Warmer*, and it is answered on the watch alongside the companion question that records the category of space they occupy at that moment. Thermal preference is the primary behavioral outcome throughout this study because it captures the

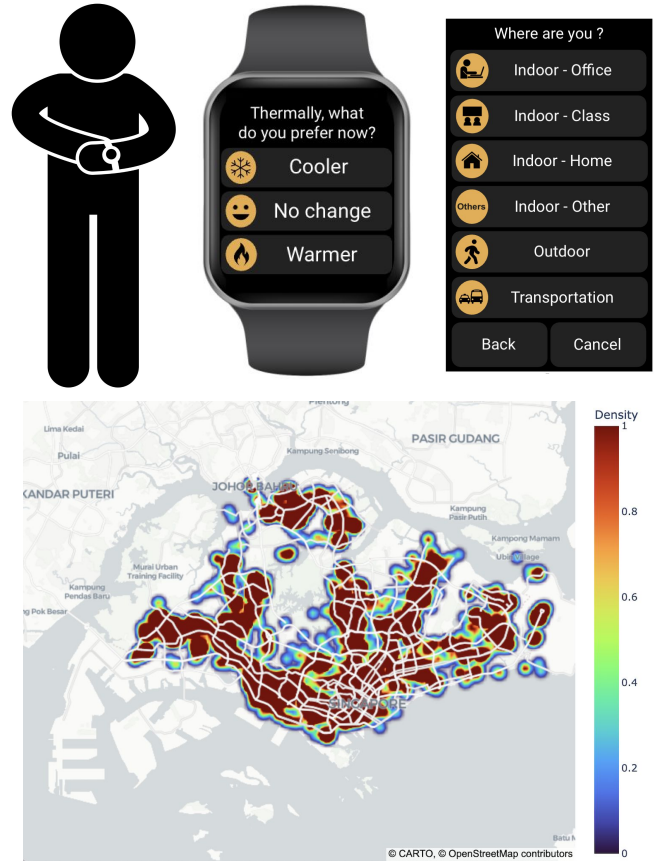


Figure 2: Cozie smartwatch micro-survey interface. The thermal-preference question offers Cooler, No change, and Warmer responses; the accompanying location question records the space category at the time of response.

desired direction of change rather than a sensation label, which maps more directly onto the adaptive actions a person might take. The location question resolves each response to an everyday space type, such as home, office, classroom, transport, and outdoor settings, so that preferences can be read in the context where they were expressed. Figure 3 summarizes the resulting survey composition, showing the overall balance of the three preference responses, how many surveys come from each space type, and how the preference mix shifts from one setting to another. To situate each response in its thermal context, each is joined to a one-hour mean outdoor heat index derived from matched local weather, which serves as an exposure proxy for the ambient conditions around the participant rather than a direct measurement of their individual microclimate. Because indoor spaces differ in how tightly their climate is controlled, each response is also assigned to an a-priori environmental-control context based on the likelihood that the occupied space is air-conditioned: outdoor settings are treated as having no air conditioning, home as possibly air-conditioned, offices, classrooms, and other indoor spaces as likely air-conditioned, and transport as very likely air-conditioned. This grouping makes it possible to ask whether a preference was expressed in a setting where the occupant could plausibly act on it through mechanical cooling,

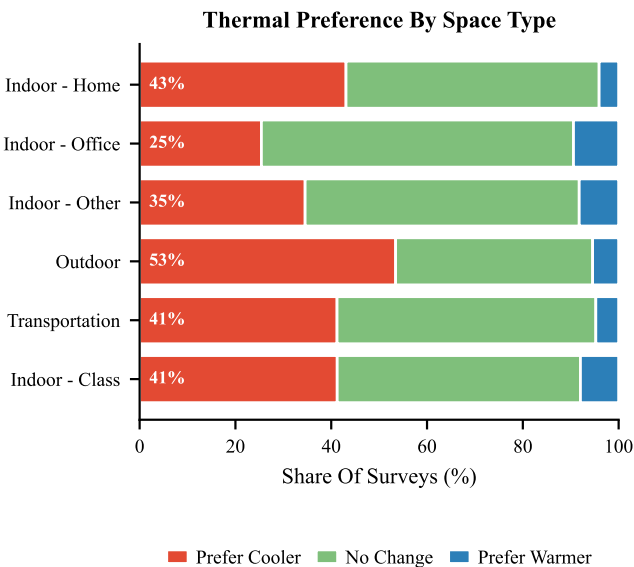
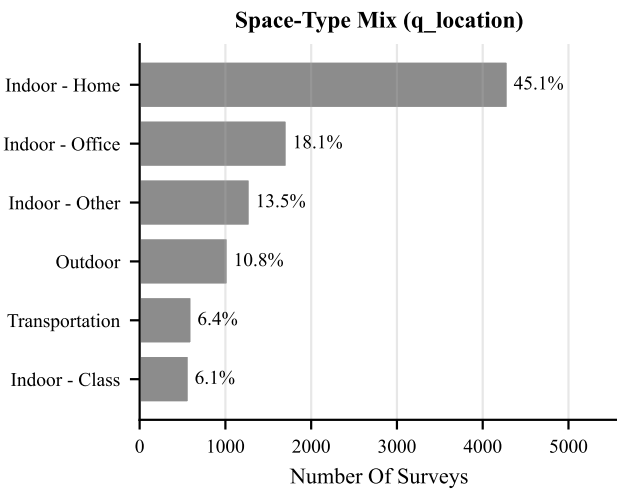
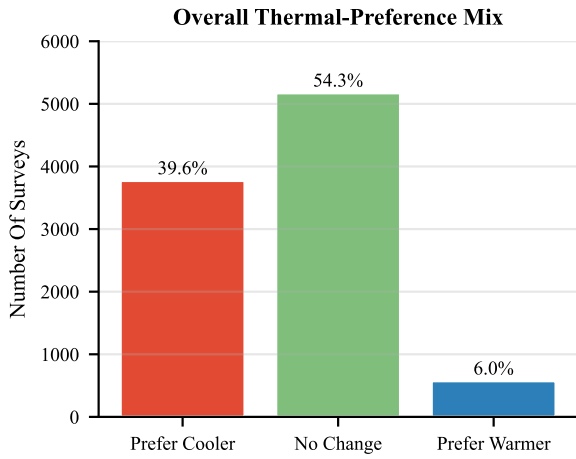


Figure 3: Survey composition and thermal preference by space type. Top panel: the overall thermal-preference mix as the number of surveys in each category (Prefer Cooler, No Change, Prefer Warmer), with each category's share of all surveys labeled above its bar. Middle panel: the number of surveys recorded in each space type, with each type's share of all surveys labeled at the end of its bar. Bottom panel: within each space type, the share of surveys indicating Prefer Cooler, No Change, or Prefer Warmer, drawn as a fully stacked bar with the Prefer Cooler share labeled inside each bar.

and it provides the control-context axis used in the mechanistic analysis that follows.

2.2. Participant eligibility and person-level features

Clustering participants requires a stable behavioral signal from each person, so the survey-level dataset is first restricted to participants who contributed enough responses on both sides of the heat range. Every response is labeled hot-side or cool-side relative to the cohort median of the one-hour outdoor heat index, which falls at 34.1 °C, so that each participant can be summarized by how their preferences differ between warmer and cooler exposures. A participant qualifies for the analysis if they submitted at least 15 responses in total, with at least 5 on the hot side and at least 5 on the cool side, applied across all locations rather than the home alone. This all-location screen retains 104 of the 106 participants, or 98%, dropping only the two who lacked sufficient coverage on one side of the split.

For each qualifying participant, three person-level features are constructed that summarize their thermal-preference behavior. The first two are the rates at which they prefer to be cooler and prefer to be warmer, computed across all of their responses and stabilized with empirical-Bayes Beta-Binomial shrinkage, which fits a cohort-wide Beta prior by marginal maximum likelihood and pulls each participant's rate toward that prior in proportion to how few responses they contributed. This shrinkage prevents participants with sparse records from producing extreme rates that would otherwise dominate the clustering. Figure 4 shows the fitted cohort priors and how the shrinkage pulls each participant's raw rates toward those priors. The third feature is a within-participant heat sensitivity, defined as the difference between their prefer-cooler rate on the hot side and on the cool side, $P(\text{Cooler} \mid \text{hot}) - P(\text{Cooler} \mid \text{cool})$, and left unshrunk so that it reflects each person's raw responsiveness to outdoor heat. Together these features place each participant in a compact behavioral space of how much they seek cooling, how much they seek warmth, and how strongly their cooling demand tracks outdoor heat, which the next subsection uses to discover the archetypes.

2.3. Archetype discovery, interpretation and scrutiny analysis

With each qualifying participant reduced to the three behavioral features, the features are standardized to zero mean and unit variance so that no single rate dominates the distance metric, and participants are grouped with k-means clustering. To choose the number of archetypes without over-fitting, candidate solutions from two to six clusters are swept and evaluated by their mean silhouette width, using many restarts and a fixed random seed for determinism. The number of archetypes to carry forward is selected by balancing silhouette cohesion against interpretability, rather than treating the silhouette optimum as definitive. The resulting groups are then interpreted through their standardized feature centroids and a principal-component projection of the feature space, alongside per-cluster silhouette widths that indicate how internally coherent each group is. Because the underlying features vary continuously, the groups are treated as descriptive behavioral summaries rather than categorical types, and membership is expected to be soft near

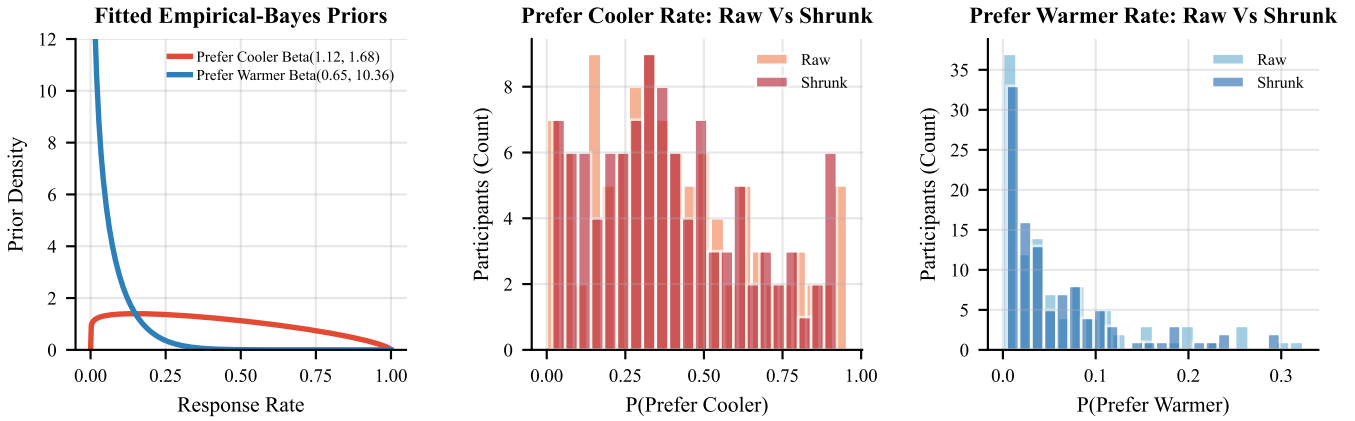


Figure 4: Empirical-Bayes feature construction. The left panel shows fitted Beta priors for prefer-cooler and prefer-warmer response rates; the center and right panels compare participants’ raw and shrunk rates for the two features.

the boundaries between them. The number of archetypes carried forward, their standardized signatures, and their descriptive names are presented in the Results.

Beyond discovering the archetypes, they are subjected to a mechanistic scrutiny that tests whether each group behaves as its signature implies. The dose-response of thermal preference to outdoor heat is examined by binning responses across the observed heat range, and preferences are compared across the a-priori environmental-control contexts, focusing on the gradient between likely air-conditioned and outdoor settings. These contrasts are quantified with distribution-free effect sizes, principally Cliff’s delta, and uncertainty is attached with participant-level bootstrap confidence intervals rather than relying on p-values. The stability of the solution is probed by re-clustering across random seeds and subsamples and by confirming that membership agrees under alternative outdoor-heat thresholds. Because survey observations are nested within participants and were not experimentally assigned, every comparison is treated as descriptive and associational rather than causal.

3. Results

Applying the person-level feature construction and clustering to the 104 eligible participants yields four behavioral thermal-preference archetypes, summarized in Figure 5. The four groups, named the Craver, the Captive, the Conventional, and the Carefree, differ in how often they seek cooling, how often they report being too cold, and how strongly their preferences track outdoor heat.

Clustering the standardized prefer-cooler rate, prefer-warmer rate, and heat sensitivity with k-means produced coherent groups across the candidate range, with the mean silhouette width highest at three clusters (0.322) and nearly identical at four (0.316). The four-cluster solution is carried forward because it separates an interpretively distinct group that the three-cluster solution merges, at negligible cost to cohesion, and it is not treated as an objectively optimal partition. Figure 6

shows that the four groups are well separated in the principal-component projection of the feature space and internally coherent, with positive mean silhouette widths in every group.

Figure 5 summarizes how the four archetypes differ along the three behavioral features, and Figure 7 quantifies these differences as standardized signatures. The Craver, or Constant Cool-Seekers ($n = 23$, 22.1%), prefers cooler conditions in roughly 74% of responses and carries a below-average heat sensitivity, seeking cooling whether or not outdoor heat is high. The Captive, or Often Over-Cooled ($n = 12$, 11.5%), stands apart through a much higher prefer-warmer rate, near 21% against only a few percent in the other groups, indicating that this group reports being too cold far more often than any other. The Conventional, or Normatively Responsive ($n = 35$, 33.7%), the largest group, shows the strongest heat sensitivity, so its prefer-cooler response rises and falls with outdoor heat in the textbook thermoregulatory pattern. The Carefree, or Heat-Unbothered ($n = 34$, 32.7%), sits below the cohort mean on all three features and reports no change in roughly 77% of responses, remaining comfortable even outdoors at comparable heat exposure.

3.1. Heat dose-response and environmental-control context

The four archetypes were derived from behavior alone, so a mechanistic scrutiny examines whether each group behaves as its signature implies once responses are placed back in their thermal and environmental context. The scrutiny proceeds through a coherent sequence: how far each group’s cooling demand is relieved as spaces become more likely to be air-conditioned, the anatomy of over-cooling in the most conditioned settings, the dose-response of preference to outdoor heat, and the balance between heat tolerance and heat exposure. Throughout, uncertainty is reported as 95% cluster-bootstrap confidence intervals that resample whole participants, and group contrasts are summarized with distribution-free effect sizes, so that no comparison rests on participant-nested responses being treated as independent. Because participants

Four Archetypes of Thermal Preference (n = 104)

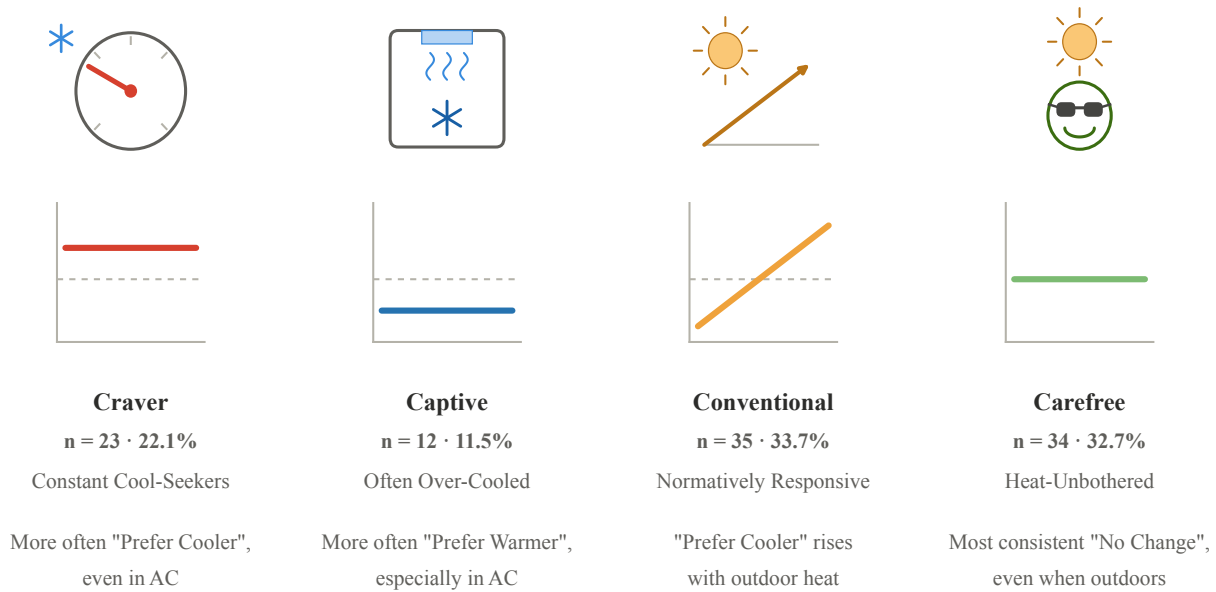


Figure 5: Overview of the four behavioral thermal-preference archetypes derived in this study (n = 104 participants). From left to right: the Craver (Constant Cool-Seekers; n = 23, 22.1%), who prefers cooler more often, even in air-conditioned spaces; the Captive (Often Over-Cooled; n = 12, 11.5%), who prefers warmer more often, especially in air-conditioned spaces; the Conventional (Normatively Responsive; n = 35, 33.7%), whose prefer-cooler response rises with outdoor heat; and the Carefree (Heat-Unbothered; n = 34, 32.7%), who most consistently reports no change, even outdoors. Each column shows a representative icon, a schematic line sketch of the archetype's thermal-preference behavior (axes unlabeled), its name and subtitle, its sample size and share of participants, and a one-line behavioral summary.

were not assigned to contexts or heat levels, every result is read as descriptive and associational rather than causal.

Figure 8 shows the share of prefer-cooler responses for each archetype across five settings ordered by their air-conditioning likelihood, from outdoors with no air conditioning to offices, classrooms, and transport that are likely air-conditioned. For three of the four groups the prefer-cooler share falls as spaces become more likely to be air-conditioned, the signature of relief from mechanical cooling: the Conventional drops from about 62% outdoors to 33% in offices and classrooms, the Captive from 53% to 23%, and the Carefree, already the most heat-tolerant, from 32% to 11%. The Craver is the exception, holding a high prefer-cooler share of roughly 64 to 82% across every context, including the spaces most likely to be air-conditioned, which points to a cooling demand that mechanical conditioning does not satisfy. The confidence intervals are wider for the smaller groups, particularly the Captive, but the separation of the Craver above the other archetypes in the likely air-conditioned settings is consistent.

Figure 9 isolates the too-cold response by plotting the prefer-warmer share of each archetype across the control contexts. The Captive is the only group that reports being too cold with any regularity, preferring warmer conditions in about 20 to 28% of responses depending on the setting, against roughly 1 to 7% for every other archetype. Its over-cooling is most pronounced in the likely air-conditioned offices, classrooms, and other indoor spaces, where the prefer-warmer share reaches

28%. The within-Captive gradient from outdoors to these likely air-conditioned settings is positive, at about 7 percentage points, but its 95% cluster-bootstrap interval is wide and includes zero, so the anatomy of over-cooling is clear in level while its context gradient stays uncertain in this small group of twelve participants.

Figure 10 turns to the dose-response of cooling demand, tracing the prefer-cooler share against outdoor heat across sextiles within three control supergroups. Only the Conventional shows a clear upward slope, with its prefer-cooler share rising by about 18 percentage points between its coolest and hottest responses (95% cluster-bootstrap interval 11 to 25 points), the signature of a group whose comfort tracks the weather. The Craver stays high and nearly flat, the Carefree stays low and nearly flat, and the Captive is flat as well, with heat couplings that are statistically indistinguishable from zero. The upward slope is clearest in the outdoor supergroup, where outdoor heat varies most and no mechanical cooling intervenes.

3.2. Robustness and boundary membership

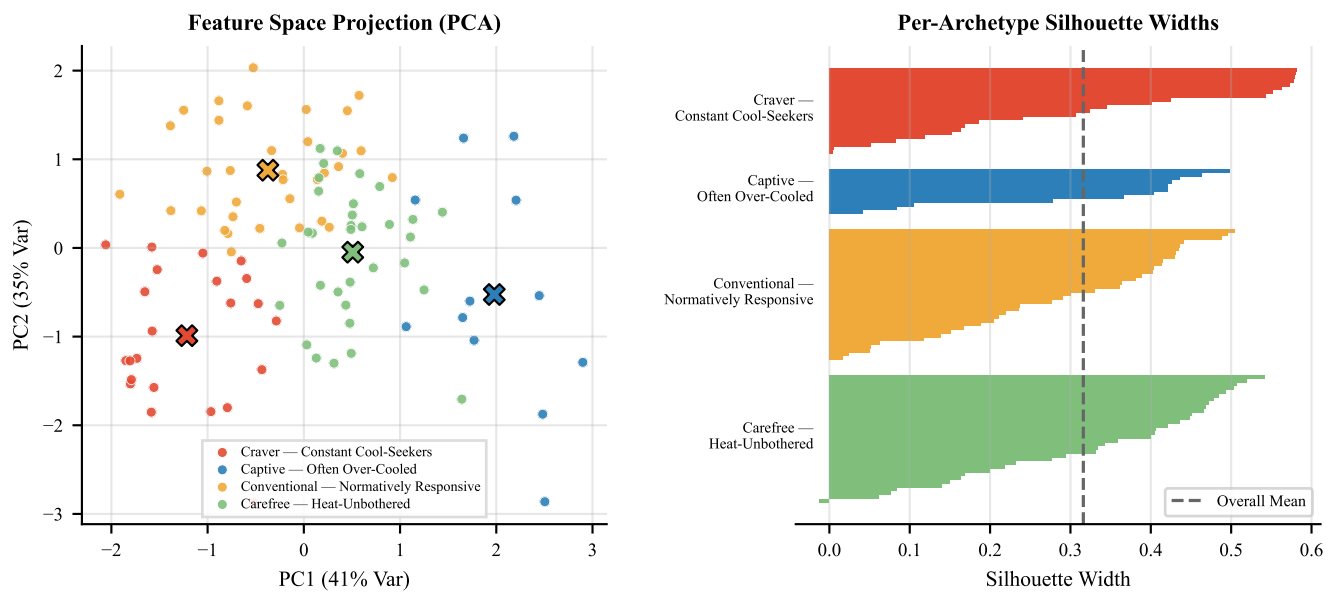


Figure 6: Four-archetype clustering diagnostics. The left panel projects the standardized feature space onto its first two principal components; the right panel shows silhouette widths for individual participants, grouped by archetype.

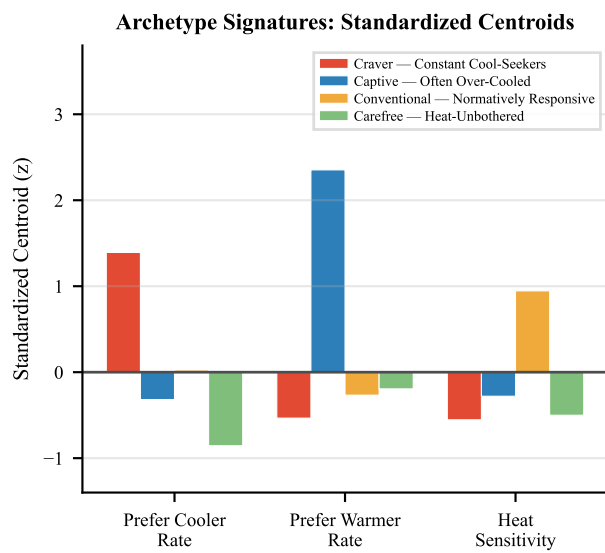


Figure 7: Standardized feature signatures of the four behavioral thermal archetypes. Lines show cluster centroids for prefer-cooler rate, prefer-warmer rate, and heat sensitivity.

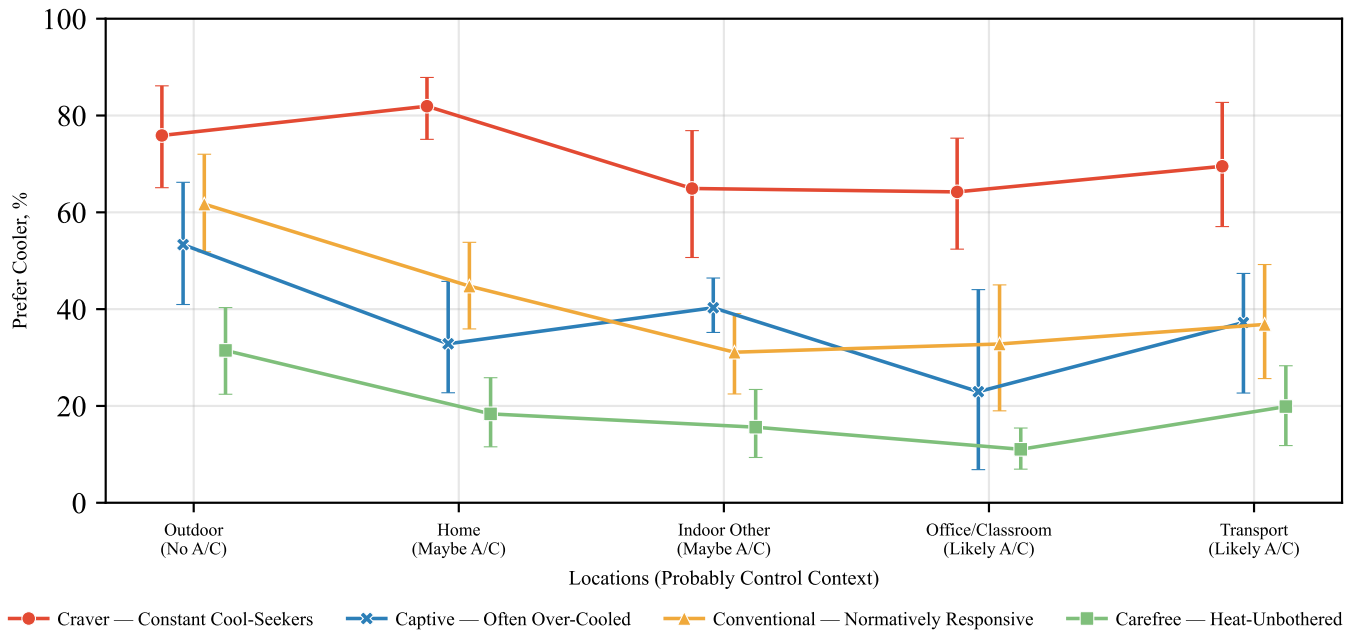


Figure 8: Prefer-cooler responses for each archetype across five probable control contexts, ordered by increasing air-conditioning likelihood (Outdoor with no A/C; Home with maybe A/C; Indoor Other with maybe A/C; Office or Classroom with likely A/C; and Transport with likely A/C), with 95 percent cluster-bootstrap confidence intervals. Each archetype uses a distinct marker in addition to color so the series remain distinguishable in black and white.

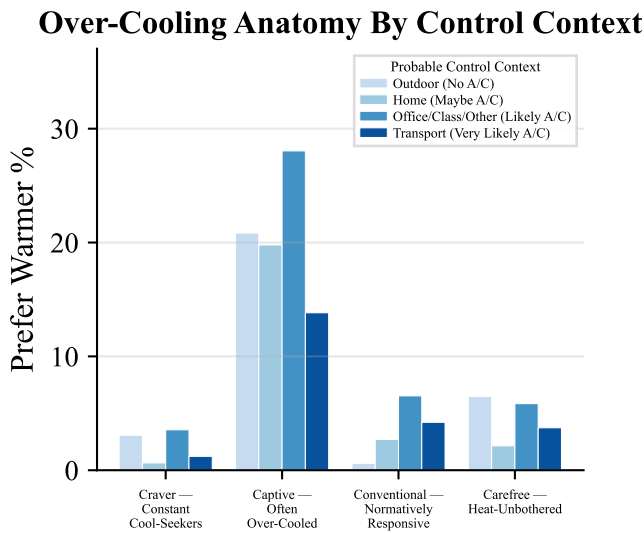


Figure 9: Prefer-warmer (too cold) responses by archetype across the four probable control contexts, ordered by increasing air-conditioning likelihood (Outdoor with no A/C; Home with maybe A/C; Office, Class, or Other indoor with likely A/C; and Transport with very likely A/C). Bars show the percentage of responses indicating a preference for warmer conditions.

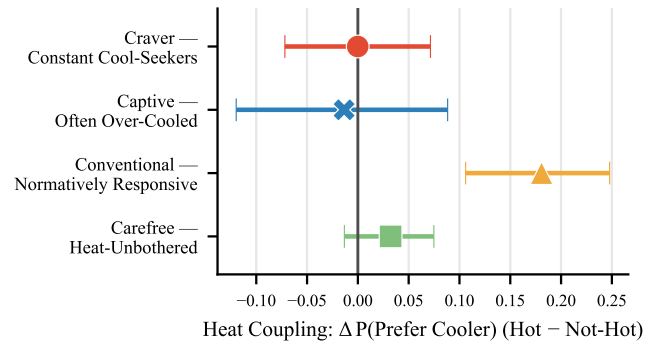
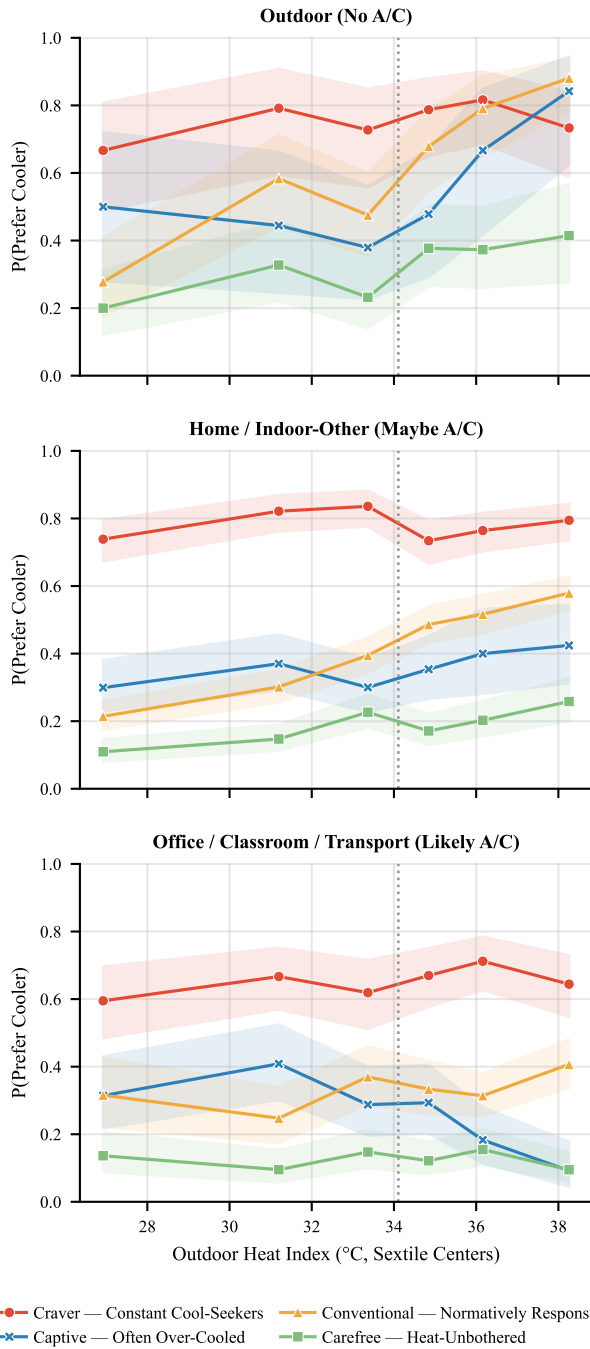


Figure 11: Heat coupling by archetype. Points and horizontal intervals show the difference in participants' prefer-cooler probability between hot and not-hot observations, with 95% bootstrap confidence intervals.

Figure 10: Probability of preferring cooler conditions versus outdoor heat, shown separately for three probable control supergroups ordered by increasing air-conditioning likelihood (top to bottom: Outdoor with no A/C; Home or Other indoor with maybe A/C; and Office, Classroom, or Transport with likely A/C). Within each panel, one line per archetype gives the prefer-cooler share across outdoor heat-index sextile centers, with Wilson 95 percent intervals as shaded bands and a distinct marker per archetype (Craver, circle; Captive, cross; Conventional, triangle; Carefree, square) so the series remain distinguishable in black and white. The dotted vertical line marks the cohort median heat index.

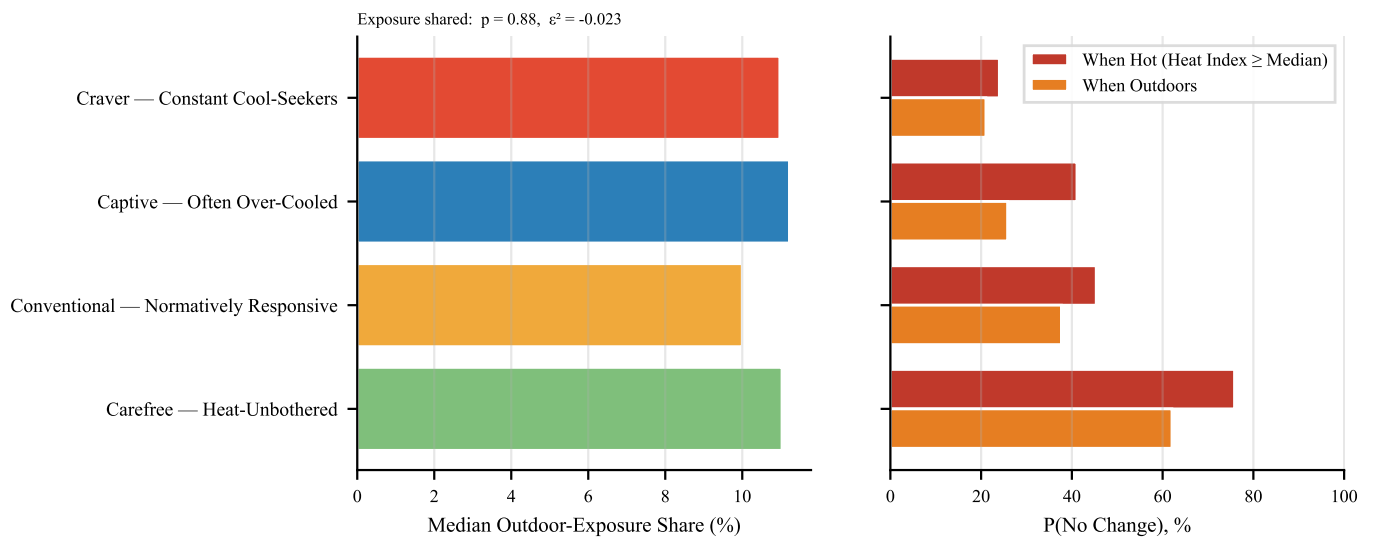


Figure 12: Outdoor exposure and provoked comfort by archetype. The left panel summarizes participants' outdoor-exposure share; the right panel shows the probability of reporting No Change when observations are both hot and outdoors.

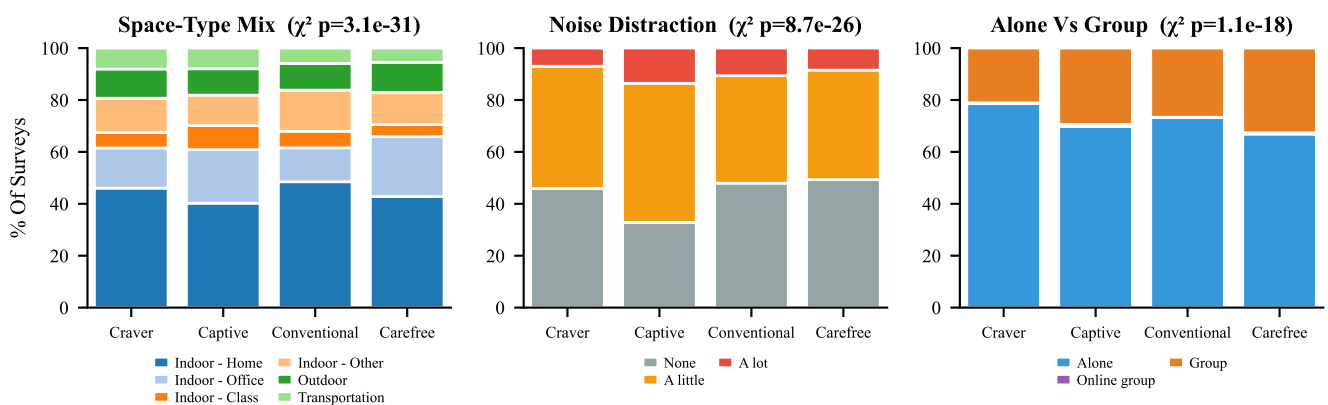


Figure 13: Survey characteristics not used to construct the archetypes. From left to right, panels show the distribution of space type, noise distraction, and alone-versus-group context across the four archetypes.

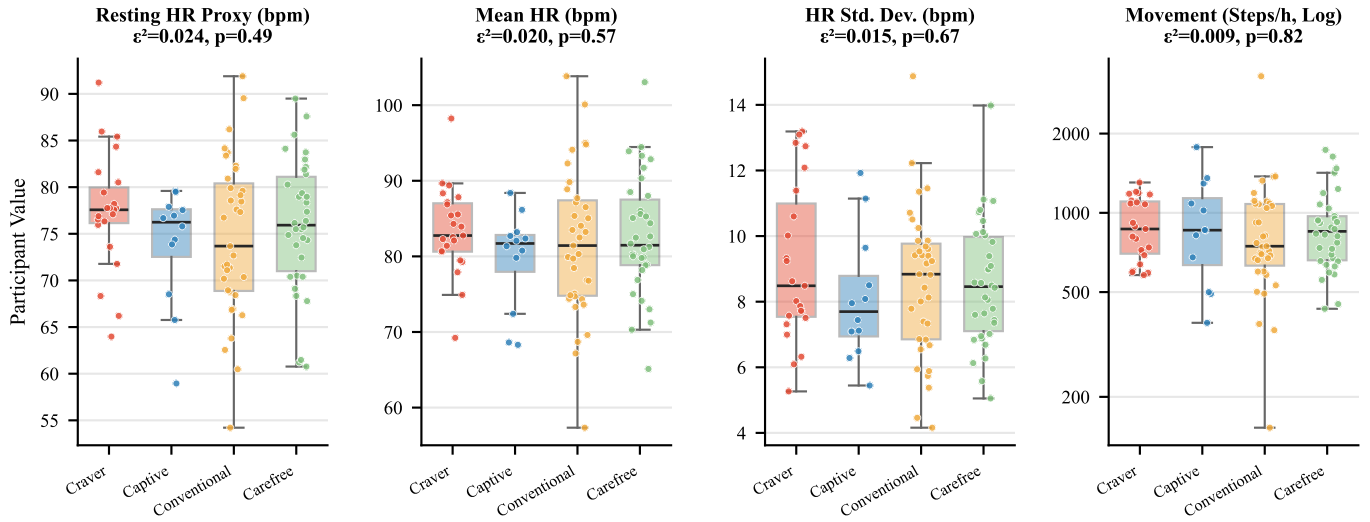


Figure 14: Wearable physiology in the one hour before each survey, summarized to one value per participant and compared across the four archetypes. From left to right: resting heart-rate proxy (mean heart rate on low-activity occasions), mean heart rate, within-hour heart-rate standard deviation (all in beats per minute), and movement (steps per hour, drawn on a logarithmic axis because a single participant is far more active). In each panel the box marks the interquartile range and median and the points are individual participants, colored by archetype. Kruskal-Wallis epsilon-squared effect sizes are all below 0.03 and none of the differences is statistically significant, so the archetypes are not distinguished by heart rate or activity.

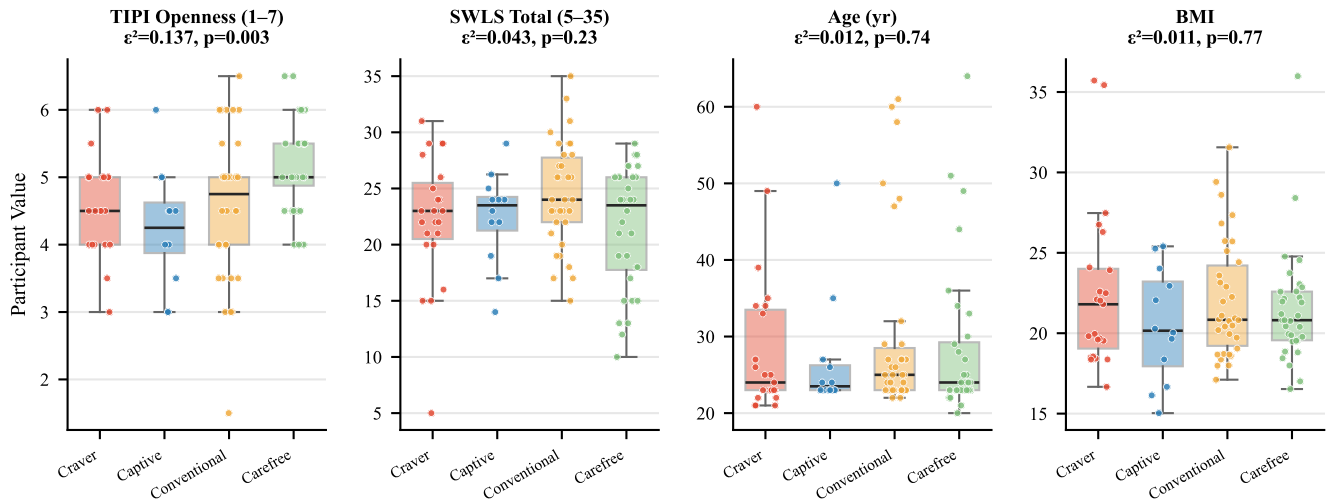


Figure 15: Onboarding measures by archetype: one candidate signal against representative null dimensions ($n = 101$ participants with onboarding data). Each panel shows a participant-level measure across the four archetypes as a box (interquartile range and median) with individual participants overlaid, colored by archetype, and is annotated with the Kruskal-Wallis epsilon-squared effect size and p-value. From left to right: Ten-Item Personality Inventory Openness (1 to 7), Satisfaction With Life Scale total (5 to 35), age in years, and body mass index. Openness is the only measure with a medium effect (epsilon-squared 0.137, $p = 0.003$), higher in the Carefree type, whereas the other dimensions overlap across archetypes.

4. Discussion

- 4.1. What the archetypes add to thermal-comfort research
- 4.2. Implications for buildings, controls, and personalization
- 4.3. Strengths and limitations
- 4.4. Future work

5. Conclusions

CRedit authorship contribution statement

CRedit authorship contribution statement

CRedit roles to be finalized by the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Data availability

The data and analysis materials will be made available in the associated public repository, subject to the privacy protections described in the accompanying data documentation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used generative AI tools to assist with organizing and polishing manuscript source files. The authors reviewed and edited all content and take full responsibility for the final manuscript.

Appendix A. Supplementary methods and robustness detail

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